

PersonalQuery Artifact Appendix

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1 Abstract

PersonalQuery is a complete reproducibility package for grounded personalized search query generation. The artifact includes the full 12-stage pipeline implementation, benchmark dataset with 134 paired queries across 10 Amazon users, comprehensive documentation, and single-command reproduction scripts. All code and data are publicly archived on Zenodo with DOI.

2 Artifact Identification

Artifact Name: PersonalQuery Reproducibility Package

DOI: 10.5281/zenodo.XXXXXX

URL: <https://zenodo.org/record/XXXXXX>

GitHub: [https://github.com/\[username\]/personalquery](https://github.com/[username]/personalquery)

License: MIT License

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3 Description

PersonalQuery implements a grounded approach to generating personalized search queries from user reviews.

Key Algorithms:

- Dual Validation: Filters preferences using review evidence + metadata verification (71.5% filtering rate)
- Iterative Style Refinement: 16-dimensional linguistic feature matching
- Spelling Difficulty Model: Cognitive difficulty scoring + user-specific error patterns

4 Requirements

Hardware: CPU 8+ cores, RAM 16GB minimum, Storage 5GB

Software: Python 3.8+, conda/pip, LLM API access

Dependencies: openai, spacy, sentence-transformers, pandas, numpy, scikit-learn

Runtime: Main results 15-30 min, Full evaluation 2-4 hours

5 Installation

1. Clone: `git clone https://github.com/[user]/personalquery.git`
2. Environment: `conda env create -f environment.yml`
3. Dependencies: `pip install -r requirements.txt`
4. API Key: `export GLM_API_KEY="your-key"`
5. Verify: `python scripts/verify-setup.py`

6 Experiment Workflow

Main Results: `python scripts/reproduce_main_results.py --user-id A130FOB1394G31`
Expected: Public 4.0, Personalized 5.8, Improvement +1.81 (15-30 min)

Full Evaluation: `python scripts/run_full_evaluation.py --all-users`
Expected: 10 users, 134 pairs, 100% positive (2-4 hours)

7 Results Reproduction

Expected Metrics:

- Public query avg: 4.0 (± 0.2)
- Personalized query avg: 5.8 (± 0.2)
- Improvement: +1.81 points (± 0.2)
- Positive rate: 100%

Tolerance: ± 0.2 accounts for LLM output variability and floating-point precision.